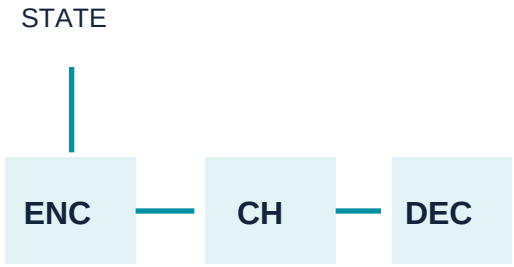
Channel Coding with State

Network Information Theory / Academic presentation / 2025

SCIENTIFIC QUESTION

How can channel state information at the encoder help mitigate interference?



KEY POINTS

- A state-dependent channel is described by P(Y|X,S).
- The encoder can use noncausal state knowledge to choose X.
- Random binning motivates the Gelfand-Pinsker coding idea.

MY CONTRIBUTION

Prepared and explained the model, random-binning principle, achievable-rate reasoning and dirty-paper-coding application.

LIMITS & ATTRIBUTION

Theoretical presentation only; no built encoder, measured channel or independently proved coding theorem.

Gelfand-Pinsker coding theorem; course presentation sources.

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